

STAND COMPOSITION AND STRUCTURE ACROSS A CHANGING HYDROLOGIC GRADIENT: JEAN LAFITTE NATIONAL PARK, LOUISIANA, USA

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Abstract: We report the results of an intensive study of forest structure and composition across a 1.4-m elevation gradient from the top of a natural levee into the backswamp of Bayou Des Familles, Jean Lafitte National Park, Jefferson Parish, Louisiana, USA. At the southernmost edge of the great bottomland hardwood forest of the Lower Mississippi Alluvial Valley (LMAV), forests of the Bayou Barataria-Des Familles distributary are undergoing rapid subsidence with resulting increased flood frequency, depth, and duration. We used data from a 4.6-ha permanently marked plot to examine patterns of distribution and regeneration in forest trees. Non-metric multidimensional scaling ordination of 23 quadrats (20 x 100 m) from this plot showed variation in forest composition across this 1.4-m elevation gradient corresponding to bottomland hardwood forest Zones III (semipermanently flooded) through V (temporarily flooded). A comparison of the size-frequency distributions of common species in upper, middle, and lower sectors of the gradient revealed deficient and poor recruitment in *Quercus virginiana*, *Acer negundo*, *Celtis laevigata*, and *Salix nigra* and episodic regeneration in *Liquidambar styraciflua*, *Taxodium distichum*, and *Quercus nuttallii*. Recruitment of the exotic species, *Sapium sebiferum*, is occurring at the low end of the gradient, as well as in canopy gaps throughout the gradient. Logistic regressions of sapling (<10 cm dbh) and tree (≥10 cm dbh) size classes as a function of elevation showed that saplings of *L. styraciflua*, *Q. nigra*, and *U. americana* occur at higher elevations than do adult trees of the same species, evidence of the rate of hydrologic change in this forest. A fourth species, *Acer rubrum*, resprouts vigorously under rising water levels and may be an effective competitor with more light-demanding, flood-tolerant species at low elevations.

Key Words: bottomland hardwood forest, logistic regression, Lower Mississippi Alluvial Valley, non-metric multidimensional scaling, regeneration, size-frequency distribution

INTRODUCTION

Bottomland hardwood forests at the southern edge of the Lower Mississippi Alluvial Valley (LMAV) are undergoing rapid hydrologic change brought about by coastal subsidence (Sasser et al. 1986, Zilkoski and Reese 1986, Conner and Day 1988, 1989). Coastal subsidence is ultimately a natural phenomenon in the upper Gulf coast—the product of sediment consolidation, oxidation of buried organic matter, changes in sediment deposition due to shifts in channel activity, and coastal down-warping under the Mississippi sediment load. However, in this area, the subsidence rate has been accelerated by oil and gas exploration, canal and levee construction, salt-water intrusion, interruption in the freshwater flow, and reduction in sediment deposition (Sasser et al. 1986, Day et al. 2000). At

Barataria Bay, at the mouth of the Barataria-Terrebonne estuary south of New Orleans, rates of relative sea-level rise are estimated at 12 mm/yr, about 2 mm/yr of which are due to eustatic global sea-level rise (Zilkoski and Reese 1986, Conner and Day 1988, Bourne 2000). Forests of the Barataria-Terrebonne estuary have been subject to increasing flood frequency, depths, and duration since the distributary ceased to carry a significant sediment load about 1700 years ago. Nevertheless, the rate of subsidence has accelerated in the last 50 years as flood patterns have been modified by human activity (Sasser et al. 1986). Global climate change, with projected shifts in rainfall and temperature regimes, is expected to have wide-spread effects on the distribution of species and the structure of natural ecosystems (e.g., Pitelka et al. 1997). In the bot-

tomland hardwood forests of the LMAV, we investigated the impact of long-term hydrologic change on population and community structure across an elevation gradient to understand better the likely consequences of one component of global change, rising sea levels, on forest ecology. In this paper, we describe the structure, composition, and regeneration of a mature bottomland hardwood forest at the southernmost edge of the great Mississippi bottomland hardwood forest complex.

Among southeastern bottomland hardwood forests, those of the LMAV are unusual in several respects. The largest continuous extent of bottomland hardwood forest in North America occurs in the LMAV (Clark and Benforado 1981, McKnight et al. 1981, Turner et al. 1981). In contrast, bottomland forests of the coastal plain often occur in narrow riparian zones fringing rivers, which traverse pine or mixed pine-hardwood upland forests. While the alluvial sediments of the Atlantic and Gulf coastal plains are derived from acid, nutrient-poor soils, sediments comprising the LMAV are derived from a large part of the continental US (McKnight et al. 1981). Alluvial soils of the lower Mississippi floodplain are often dominated by shrink-swell clays and are characterized by a weakly acid to alkaline pH, high phosphorous availability, and high cation exchange capacity (Schumacher et al. 1988). In addition, the floodplain of the lower Mississippi is topographically complex in comparison to the relatively narrow channels of most smaller coastal plain rivers (McKnight et al. 1981). Old distributaries, meander scrolls, point bars, levees, swales, and backswamps produce fine-scaled topographic relief, heterogeneous hydrologic environments, and spatially complex forests.

Here, we describe the population structures of tree species across a hydrologic gradient affected by both coastal subsidence and rising sea levels. Our gradient runs from the top of a natural levee into the back swamp of an old distributary of the Mississippi River. Specifically, we ask 1) how do forest composition and structure change as a function of elevation along this hydrologic gradient; 2) how do the size-class distributions of important tree species vary along the gradient, and 3) do distributions of sapling and tree size classes differ along the elevation gradient?

STUDY SITE

The study site is on the backslope of a natural levee of Bayou des Familles at Jean Lafitte National Park, Jefferson Parish, Louisiana, USA located at the seaward edge of the bottomland hardwood forest on the Mississippi alluvial plain, approximately 4.3 km south of New Orleans (Figure 1). The Bayou Barataria-Bay-

ou Des Familles distributaries of the Mississippi River were active from about 1500 BC to 200 AD when they carried around 30% of the river flow (Swanson 1991). Overbank flooding during this period created natural levees along the east and west banks of Bayou des Familles. Water flow in Bayou des Familles was subsequently much reduced due to the formation of sediment bars at its junction with the main channel. During historic times, the bayou continued to carry flow but no sediment until ca. 300 yr ago (Swanson 1991). Although the hydrology of most bottomlands in the LMAV have been affected by human activities, that impact has been small in the bottomland hardwoods of Jean Lafitte NP. Flood frequency, timing, duration, and depth reflect rates of rainfall (1572 mm/yr in New Orleans), evapotranspiration, and drainage into Bayou Barataria. Duration and depth of flooding are greatest during the winter months (November–March) when surface water often may top the natural levee. During the growing season, the Sharkey clay soils of the levee crests may become dry and cracked in the manner of shrink-swell clays. The backswamp of Bayou Des Familles retains a natural drainage connection to Bayou Barataria and thence to the Gulf of Mexico. Salt-water intrusion during storm surge or sustained on-shore winter winds has not been known to reach the backswamp since the establishment of the park (D. Muth, Jean Lafitte National Park, personal communication). A Jefferson Parish levee above the park has effectively reduced the size of the watershed and, therefore, the total water flow through the backswamp and into Bayou Barataria.

At the southernmost extent of the bottomland hardwood forest distribution in the LMAV, the mature bottomland hardwood forest at Jean Lafitte grades from *Quercus virginiana*-dominated forest on the natural levee ridges into *Taxodium distichum* *Fraxinus profunda*-dominated backswamps (White et al. 1983). Although bald cypress were occasionally logged from the backswamp, perhaps as recently as the 1950s, there is no evidence of agriculture or logging on the natural levee in the vicinity of the study plot (Swanson 1991). The dwarf palm, *Sabal minor* (Jacq.) Pers., is abundant in the understory.

METHODS

We established a 4.6-ha permanent mapped plot (100 m wide by 460 m long) from the top of the natural levee on Bayou des Familles down the backslope to the edge of permanent standing water (460m). A grid of 20-m $\times$ 20-m cells with one random point within each cell was surveyed on the plot using a laser theodolite (Topcon GTS-213 Electronic Total Station, Topcon Positioning Systems, Inc., Pleasanton, CA,

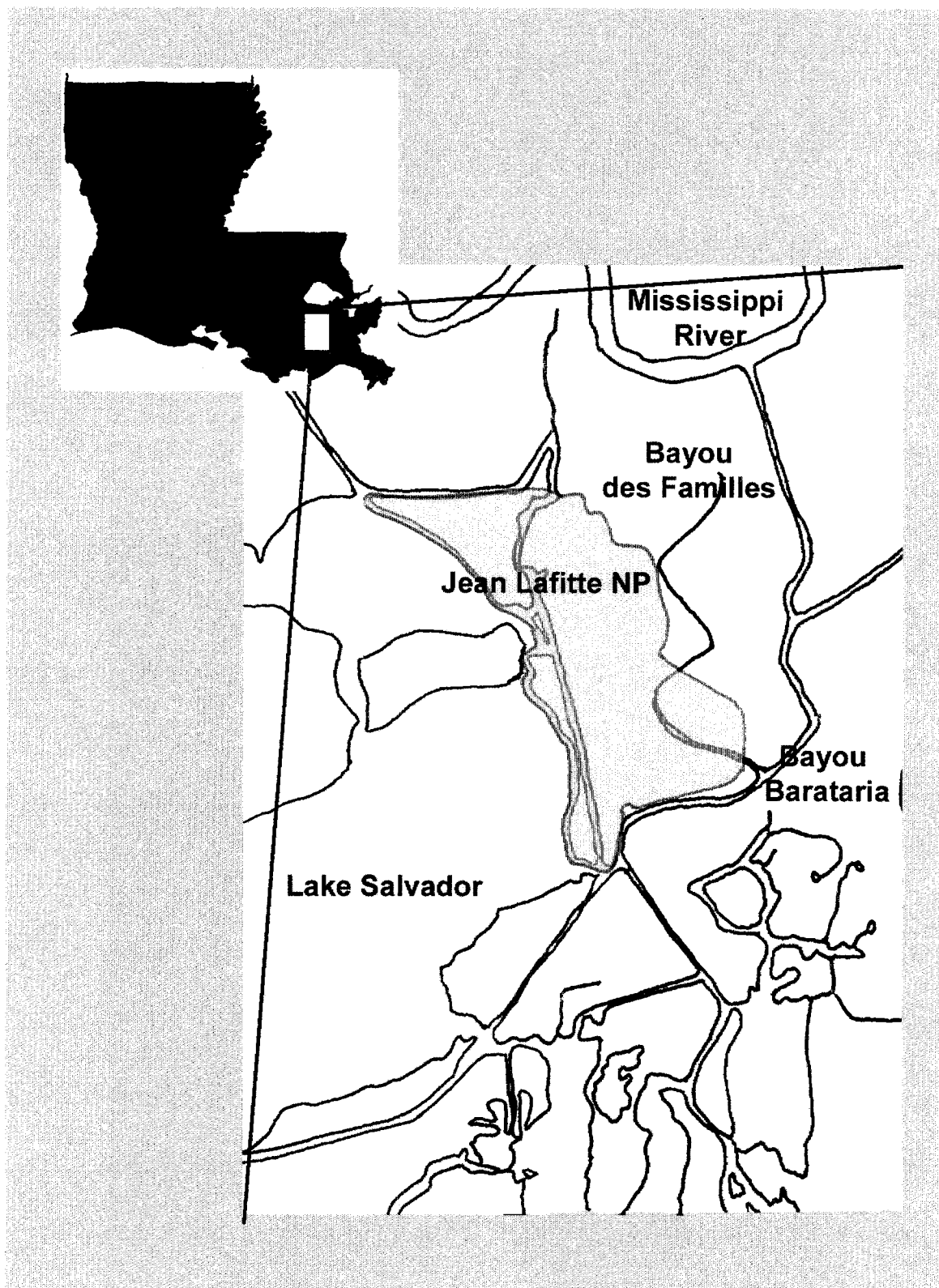


Figure 1. Location of Jean Lafitte National Historical Park and Preserve in Louisiana, USA.

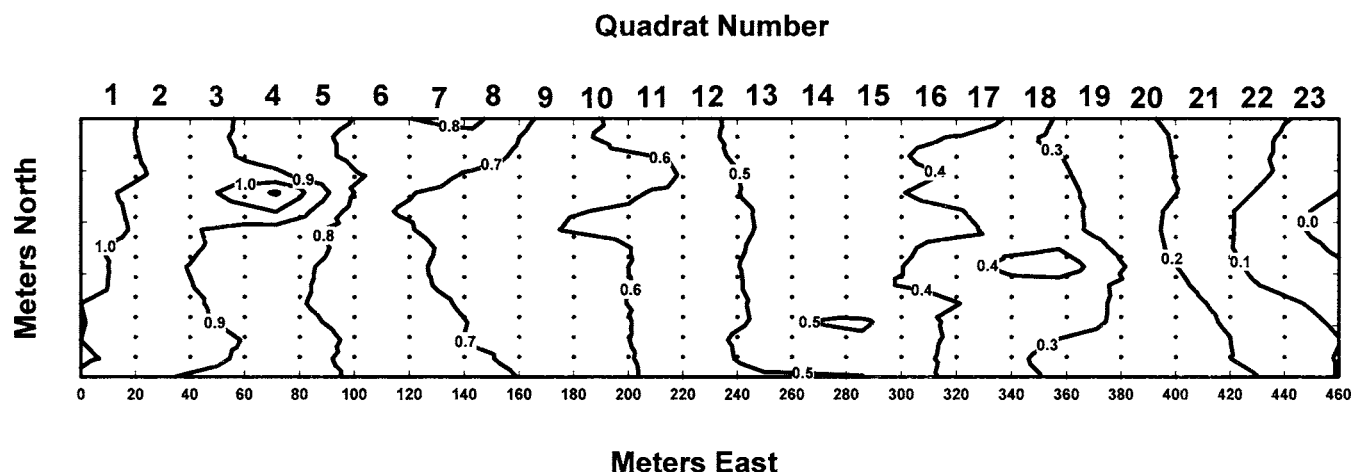


Figure 2. Topographic map based on elevations measured at grid and random points. Contour intervals are in meters above sea level. Positions of quadrats used in subsequent ordinations are indicated.

USA). All stems ≥ 2.5 cm dbh were mapped, measured (dbh), and identified. Using the Topcon Total Station, we also measured elevation (± 5 mm) at all grid and random points, as well as at the bases of all mapped stems. A topographic contour map of the plot was constructed in Sigma Plot (SPSS, Inc., San Rafael, CA, Ver. 4.0) by kriging elevations of mapped points based on grid and random points.

We divided the plot into 23 quadrats (20 m $\times$ 100 m) arrayed perpendicular to the long axis of the plot to examine change in vegetation structure across the topographic gradient. We used non-metric multidimensional scaling (NMDS), a technique that has been shown to be robust for ordination of community data (Minchin 1987). NMDS finds an ordination of n quadrats in k dimensions, such that the distances among all pairs of quadrats in the ordination are, as far as possible, in rank-order agreement with compositional dissimilarities among the quadrats. We used importance values (IV), calculated as the mean of relative density and relative dominance, to express species abundance and calculated dissimilarities using the Bray-Curtis index (Bray and Curtis 1957). The Bray-Curtis index has been shown to be one of the most effective for community ordination (Faith et al. 1987). Ordinations were performed in SAS Version 6.12 using PROC NMDS (Anon. 1997).

On the basis of this ordination, we identified three groups of quadrats based on similarities in species composition along the topographic gradient. While these groups were associated with topographic position, they also reflected species responses to hydrologic pattern that may not have been a linear function of elevation. In each of these three groups, we calculated stem size-class distributions of 12 common species to explore patterns of abundance and regeneration as a function of topographic position. We also used logistic

regression (Hosmer and Lemeshow 1989) to model species distributions as a function of elevation. Logistic regression estimates the probability of a binary event (in this case, species presence) given certain site conditions (here, elevation) (Austin et al. 1990, Battaglia 1998, Wiser et al. 1998). Elevations at the bases of all stems combined (5688 points) provided the frequency distribution of elevations. These models reflect not only the topographic distribution of species but differences in stem densities across the elevation gradient. The regression estimates the *relative* frequency of a species as a function of elevation. We used a logit link function to transform the estimator into a probability value (SAS Version 6.12, PROC GENMOD, Anon. 1997) and a likelihood ratio test to examine the significance of the deviance from the null model with the inclusion of elevation (E) and E². The residual deviance is measured as $-2 \log_e L$, where L = maximum likelihood. For four species (*A. rubrum*, *L. styraciflua*, *Q. nigra*, and *U. americana*), we also had sufficient sample sizes to compare distributions of sapling (2.5–9.9 cm dbh) and adult (>10 cm dbh) size classes.

RESULTS

The 4.6-ha plot spanned approximately 1.4 m elevation in a distance of 460 m from the top of the natural levee on Bayou des Familles to the lower plot boundary in the backswamp (Figure 2). We measured and mapped a total of 5688 stems of 23 species across this gradient.

Ordination of 23 0.2-ha quadrats of the Lafitte plot confirmed the strong influence of topographic position on vegetation structure and composition as seen in the array of quadrats numbered consecutively from the top (1) to the bottom (23) of the gradient (Figure 3). The dominant axis of the ordination (dimension 1) was

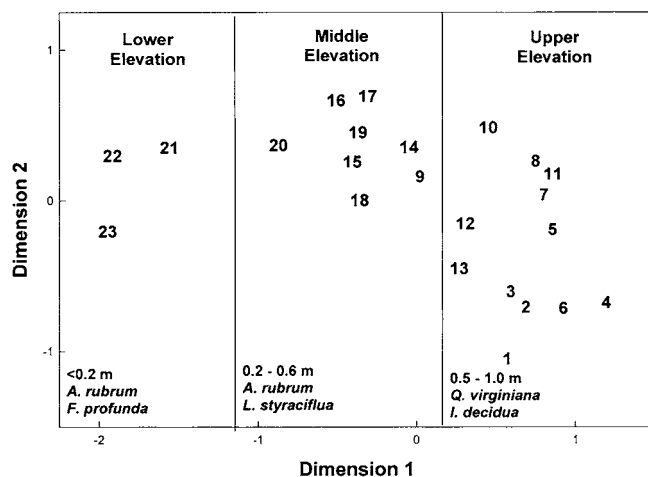


Figure 3. NMDS ordination of 23 quadrats (20 x 100 m) along an elevation gradient from the top of the natural levee into the backswamp at Jean Lafitte National Park. The ordination is based on Importance Values (relative density + relative dominance) of trees ≥ 5 cm dbh. Quadrats are numbered from 1 (top of levee) to 23 (backswamp) as depicted in Figure 2. Low, middle, and high elevation sectors of the gradient used in subsequent analyses are indicated along with their most important species.

strongly related to elevation ($r_s = -0.872$, $p < 0.01$, $df = 21$). The secondary axis revealed more heterogeneity in stand structure at the upper than the lower end of the gradient. This heterogeneity was associated with the presence or absence of large, widely spaced *Q. virginiana* trees. We divided the principle gradient into three sectors to reflect changes in species composition and dominance along the gradient. Table 1 details the composition and relative abundances of species in lower, middle, and upper elevation sectors. While total basal area (m^2/ha) was similar among the three sectors, mean stem sizes were smaller and stem densities greater at lower elevations. *Acer rubrum* was abundant across the gradient but reached an IV of more than 50 at the lowest elevation where it shared dominance with *F. profunda*. In addition to *A. rubrum*, characteristic dominants in the middle elevation sector were *L. styraciflua* and *U. americana*. In the upper elevation sector, *Q. virginiana*, by virtue of large sizes, and *I. decida*, by virtue of high densities, shared dominance. Other common species on the levee ridge included *A. rubrum*, *L. styraciflua*, *Q. nigra*, and *U. americana*.

We used these three topographic sections to explore the influence of elevation on population size structure for several common species (Figure 4). Of 15 species with densities of >15 stems/ha in at least one of the three elevation classes, two species (*C. laevigata* and *Q. virginiana*) were deficient in small sizes and may not be regenerating. We measured no stems of *Q. virginiana* smaller than 40 cm dbh. Size distributions for

three species (*L. styraciflua*, *T. distichum*, *Q. nuttallii*) suggest that recruitment may be sporadic at some elevations, perhaps associated with drought years or catastrophic opening of the canopy from windstorms. *Salix nigra* and *S. sebiferum*, two species that can show rapid growth in high light conditions, may be in this category as well. In two species (*U. americana* and *Q. nuttallii*), small sizes were most abundant at the lowest elevation, whereas larger sizes were most common at middle or upper elevations. We are not confident of our ability to distinguish *F. pennsylvanica* and *F. profunda* in small size classes and therefore do not present data on regeneration in those species. Adult *F. profunda* is most abundant at lower elevations, where it produces multiple stems from the root crown; *F. pennsylvanica* is most abundant at middle to high elevations.

Large densities of small *A. rubrum* stems were recorded at all elevations. At the low end of the gradient, sprouts from root crowns dominated the small size classes, while establishment from seed appeared to be common at upper elevations. Canopy trees of *A. rubrum* were common at both low and middle elevations but less common at the highest elevations (Figure 4).

Logistic regressions based on all size classes are given for nine species in Figure 5. Separate regressions for sapling and tree distributions of four abundant species are presented in Figure 6. These patterns reflect both the changing hydrologic regime and differences in total stem densities across the elevation gradient, which is considerable in the Lafitte plot (960 stems/ha at the upper elevation vs 2756 stems/ha at the lower elevation). Thus, *U. americana* and *L. styraciflua* saplings were smaller proportions of all stems at the lower elevation than the upper elevation. In contrast, absolute abundances of *U. americana* saplings were greatest at the lower elevation.

In all cases, addition of elevation to the null model improved the regressions significantly, as did addition of a quadratic term (Table 2). Locations of modal relative abundances for most of these species were at upper elevations. Saplings of *L. styraciflua*, *Q. nigra*, and *U. americana* are relatively more abundant at upper elevations than are trees of these species (Figure 6). Large relative abundances of *A. rubrum* saplings at low elevations are due to copious root sprouts.

DISCUSSION

Variation in Forest Structure Across a Hydrologic Gradient

Bottomland hardwood forests are notable for fine-scaled spatial pattern produced by steep hydrologic gradients (Brinson 1990, Taylor et al. 1990, Sharitz

Table 1. Composition and structure of forest at Jean Lafitte National Park. Importance values are calculated as the means of relative density and relative dominance on stems >5 cm dbh. DBH in cm.

	Flood Tolerance ¹	Lower Elevation				Middle Elevation				Upper Elevation			
		Mean dbh	Trees/ha	m ² /ha	IV	Mean dbh	Trees/ha	m ² /ha	IV	Mean dbh	Trees/ha	m ² /ha	IV
<i>Acer negundo</i> L.	moderate					14.7	34	0.65	2.3	16.3	87	2.08	7.5
<i>Acer rubrum</i> L.	moderate	7.2	1704	14.08	51.8	10.4	618	10.34	39.6	10.3	161	2.33	11.8
<i>Carya aquatica</i> (Michx.) Nutt.	moderate	6.0	2	<0.01	<0.1	15.0	9	0.23	0.7	21.9	<1	0.02	0.04
<i>Celtis laevigata</i> Willd.	moderate to tolerant					31.1	11	1.03	2.0	32.6	18	1.72	3.4
<i>Carpinus caroliniana</i> Walt.	weak									7.3	3	0.01	0.2
<i>Cornus drummondii</i> C. A. Meyer	tolerant					5.6	3	0.01	0.1	5.6	9	0.02	0.5
<i>Crataegus viridis</i> L.	moderate	7.3	4	0.02	0.1	8.3	19	0.12	0.9	7.5	31	0.16	1.9
<i>Fraxinus pennsylvanica</i> Marsh.	moderate	14.4	64	1.35	3.2	16.5	63	2.29	5.8	18.2	54	2.04	5.8
<i>Fraxinus profunda</i> (Bush) Bush	tolerant	11.0	692	9.66	26.9	7.1	60	0.35	2.9	7.0	3	0.02	0.2
<i>Ilex decidua</i> Walt.	moderate	4.0	20	0.03	0.4	4.8	202	0.42	8.6	4.8	286	0.59	15.7
<i>Liquidambar styraciflua</i> L.	moderate	14.9	40	1.07	2.3	36.9	63	8.76	15.2	26.0	68	6.20	12.4
<i>Nyssa aquatica</i> L.	tolerant	8.0	4	0.02	0.1								
<i>Quercus laurifolia</i> Michx.	moderate to weak	15.2	28	0.73	1.6	16.6	26	1.14	2.7	26.4	3	0.27	0.5
<i>Quercus nigra</i> L.	moderate to weak	11.1	12	0.19	0.5	10.1	43	0.63	2.6	12.8	101	3.32	10.0
<i>Quercus nuttallii</i> Palmer	moderate	11.9	20	0.30	0.8	44.6	14	3.17	5.2	51.7	8	2.05	3.4
<i>Quercus virginiana</i> Mill.	weak to intolerant					44.8	3	0.67	1.1	98.4	11	9.82	14.7
<i>Salix nigra</i> Marsh.	tolerant	24.3	54	2.78	5.1								
<i>Sapinum sebiferum</i> (L.) Roxb.	moderate	12.6	22	0.33	0.9								
<i>Taxodium distichum</i> (L.) Rich.	tolerant	23.9	28	2.97	4.9	41.7	1	0.09	0.2				
<i>Ulmus americanum</i> L.	moderate	4.7	60	0.15	1.3	22.8	84	4.57	10.0	17.2	114	4.00	11.7
Other species			2	0.01			1	0.04			3	0.12	
Total			2756	33.69			1254	34.51			960	34.77	

¹ Flood tolerance categories from McKnight et al. (1981) except *S. sebiferum* from Jones and Sharitz (1990) and Conner (1994).

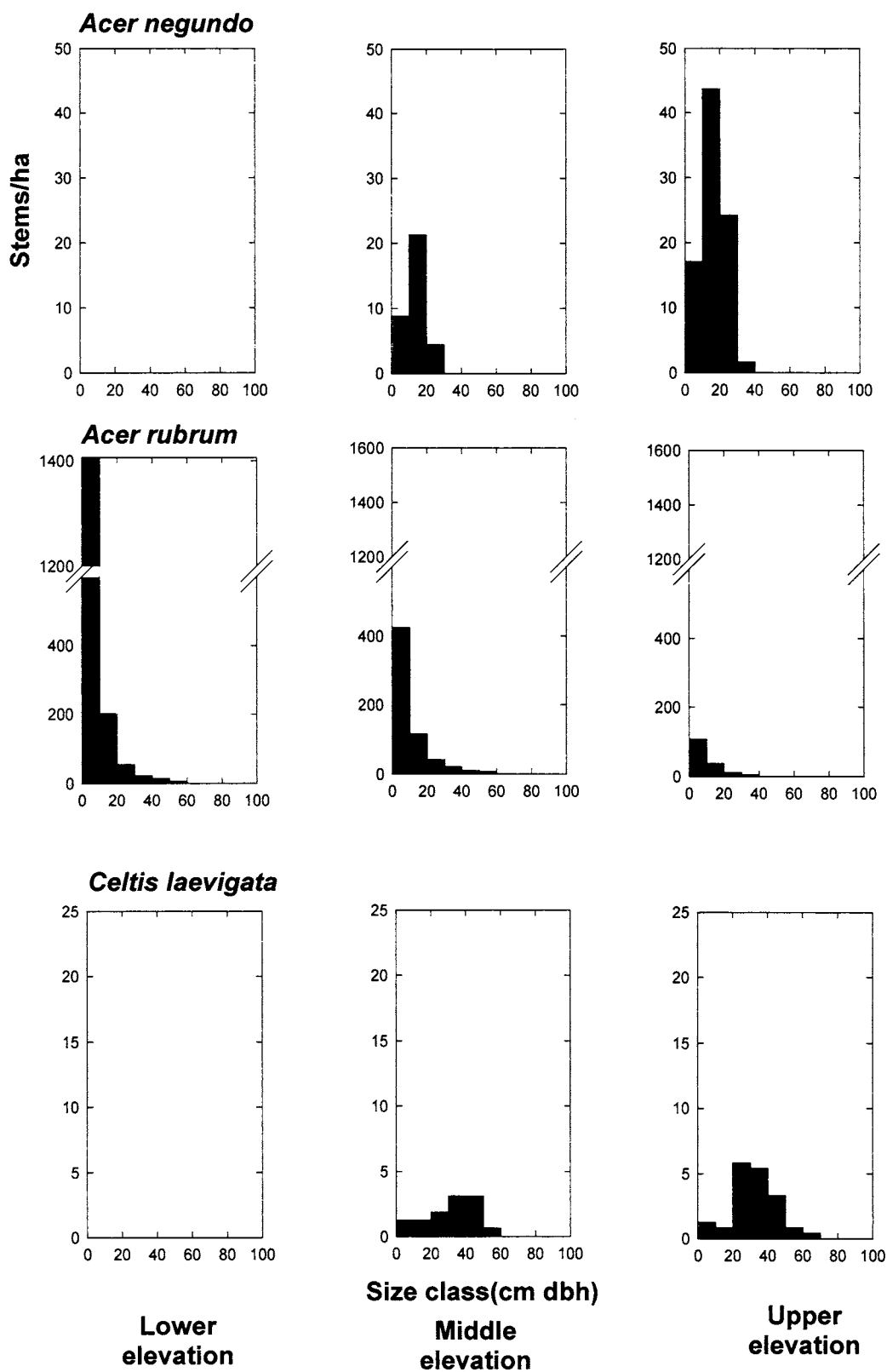


Figure 4. Size-class distributions (stems/ha) of common trees ≥ 2.5 cm dbh in quadrats in 3 elevation zones (lower: <0.2 masl; middle: $0.2\text{--}0.6$ masl; upper: $0.5\text{--}1.0$ masl). Axis scales differ among species but not among elevations.

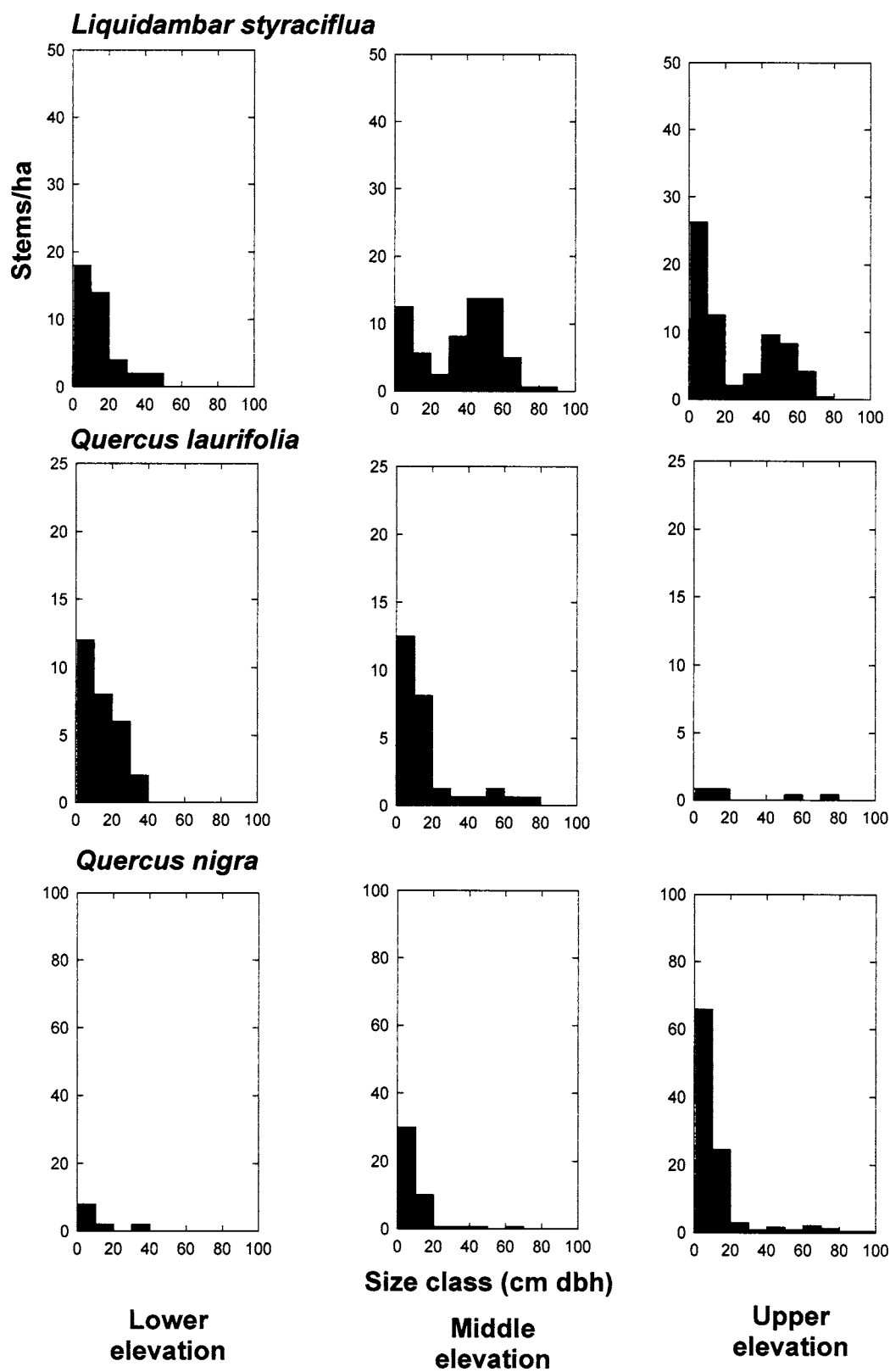


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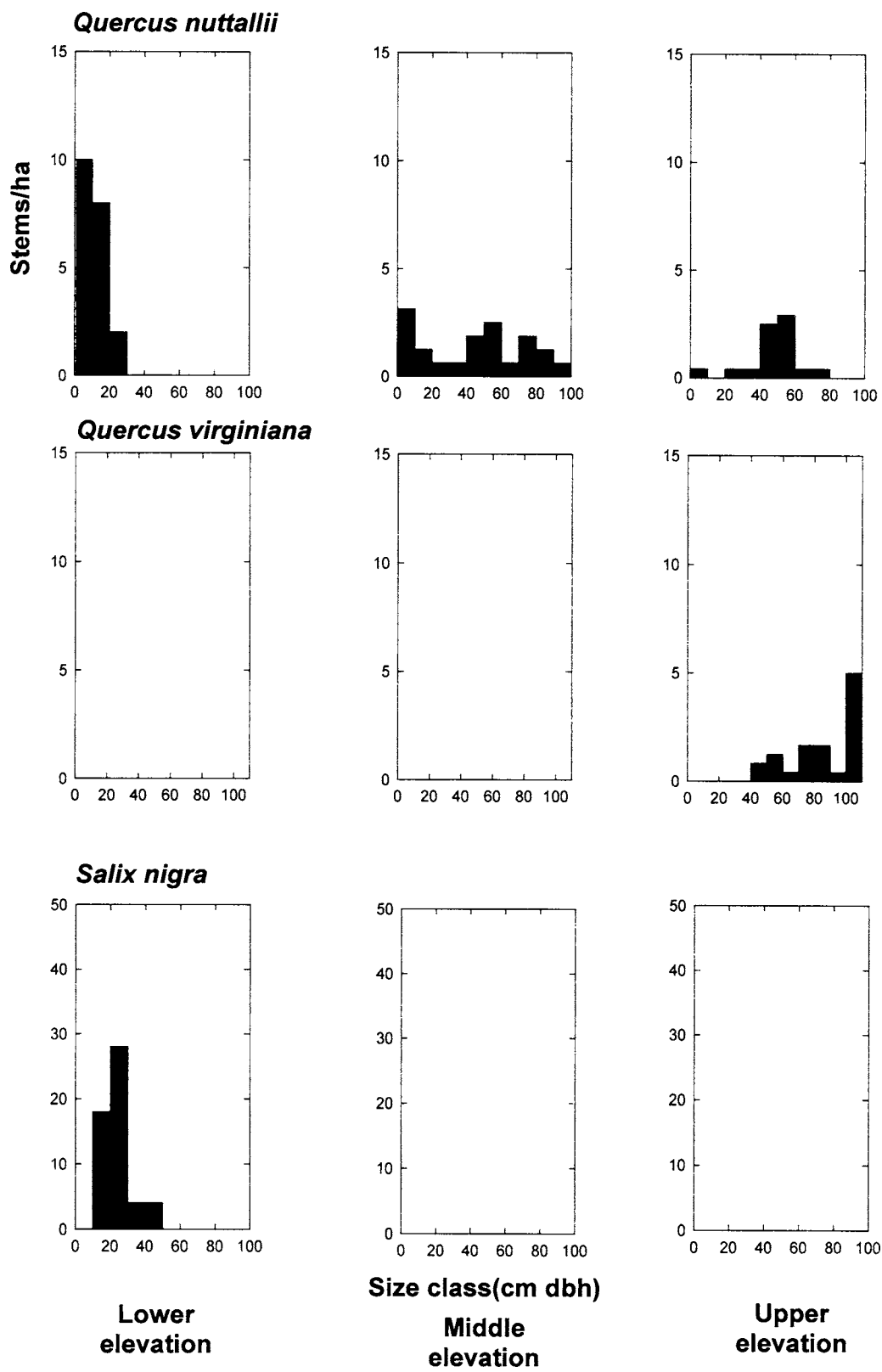


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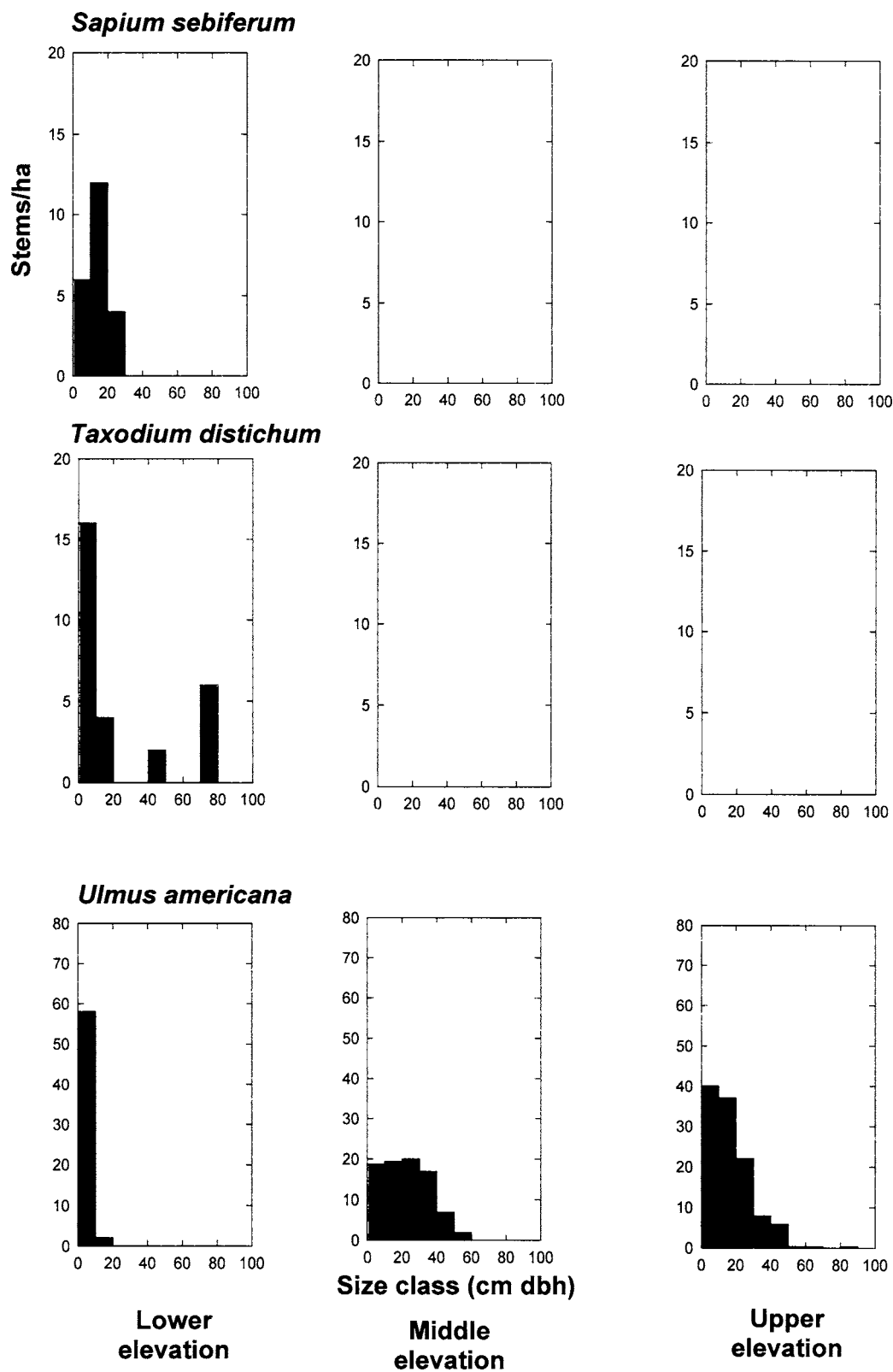


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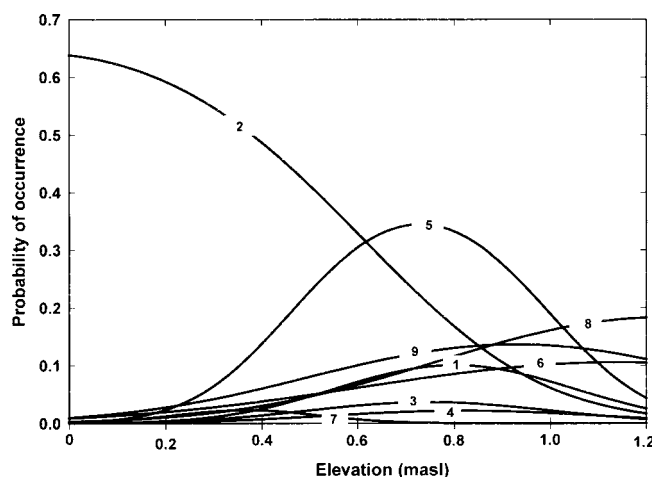


Figure 5. Probability response curves for stems ≥ 2.5 cm dbh as a function of elevation. Curves are based on logistic regressions. 1 = *A. negundo*; 2 = *A. rubrum*; 3 = *C. laevigata*; 4 = *C. viridis*; 5 = *I. decida*; 6 = *L. styraciflua*; 7 = *Q. laurifolia*; 8 = *Q. nigra*; 9 = *U. americana*. *Fraxinus* spp. are not included because of uncertainties in identifications of juveniles.

and Mitsch 1993). Across a 1.4-m elevation gradient at Lafitte, our plot traverses communities described by Clark and Benforado (1981), Wharton et al. (1982), Taylor et al. (1990), and Mitsch and Gosselink (1993) as Zones III (semi-permanently flooded) through V (temporarily flooded). While basal area/ha was similar among the three elevation sectors, mean stem sizes were smaller and stem densities greater at lower elevations. In contrast to studies of other elevation gradients (summarized by Brinson 1990), we found that species richness of trees did not change appreciably as a function of the hydrologic gradient; however, our gradient did not include Zone II (permanently flooded) communities, which are usually dominated by one or two species only.

The distributions of species across the elevation gradient were generally consistent with the flood tolerance classification suggested by McKnight et al. (1981) and others. Most of the species at Lafitte were classified as tolerant or moderately tolerant. Less flood-tolerant species—*Q. virginiana*, *Carpinus caroliniana*, and *C. laevigata*—are rare or declining members of the community. *Quercus virginiana* at Lafitte is clearly a senescent population, and *C. laevigata* likely is declin-

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Figure 6. Probability response curves of trees (≥ 10 cm dbh) and saplings (2.5–9.9 cm dbh) of 4 abundant species as a function of relative elevation. Curves are based on logistic regressions (Table 2). a. *A. rubrum*; b. *L. styraciflua*; c. *Q. nigra*; d. *U. americana*.

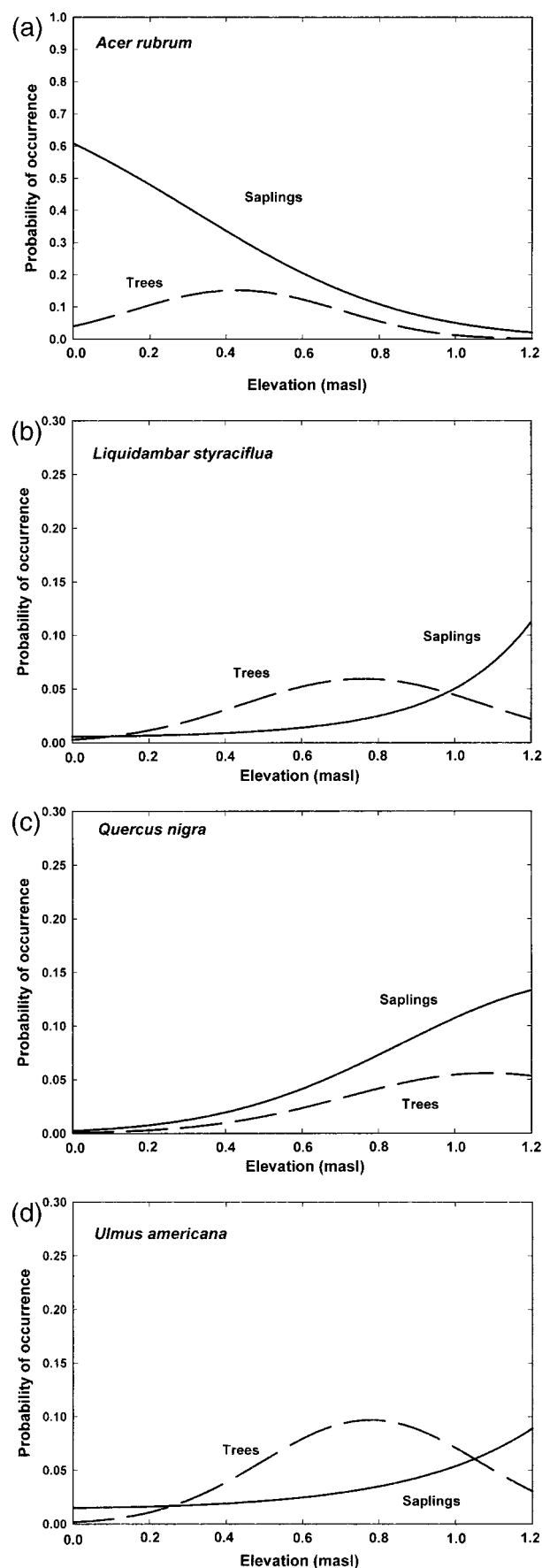


Table 2. Likelihood ratio statistics for successive models (linear and quadratic models) in logistic regressions of tree distribution as a function of elevation. Associated probability values based on asymptotic chi-square distributions are for the significance of inclusion of each additional parameter (PROC GENMOD, Anon. 1997). Regressions are plotted in Figures 5 and 6.

Species	Linear Model	p	Quadratic Model	p
<i>Acer rubrum</i> saplings	169.661	<0.001	4.764	<0.05
<i>Acer rubrum</i> trees	29.514	<0.001	37.771	<0.001
<i>Acer rubrum</i> all stems	727.640	<0.001	32.382	<0.001
<i>Acer negundo</i> all stems	405.954	<0.001	194.820	<0.001
<i>Celtis laevigata</i> all stems	14.765	<0.001	7.497	<0.01
<i>Crataegus viridis</i> all stems	186.731	<0.001	140.266	<0.001
<i>Ilex decidua</i> all stems	447.204	<0.001	383.769	<0.001
<i>Liquidambar styraciflua</i> saplings	337.918	<0.001	16.541	<0.001
<i>Liquidambar styraciflua</i> trees	141.070	<0.001	80.728	<0.001
<i>Liquidambar styraciflua</i> all stems	233.866	<0.001	15.281	<0.001
<i>Quercus laurifolia</i> all stems	150.207	<0.001	320.142	<0.001
<i>Quercus nigra</i> saplings	508.343	<0.001	17.974	<0.001
<i>Quercus nigra</i> trees	449.600	<0.001	41.701	<0.001
<i>Quercus nigra</i> all stems	150.207	<0.001	320.142	<0.001
<i>Ulmus americana</i> saplings	114.674	<0.001	6.477	<0.05
<i>Ulmus americana</i> trees	250.131	<0.001	131.151	<0.001
<i>Ulmus americana</i> all stems	66.110	<0.001	35.416	<0.001

ing. *Carpinus caroliniana* is scarce. Both rising water levels and heavy clay Sharkey soils likely have contributed to the low abundance and reproduction of these species.

A large proportion of the smallest size classes at the lower elevations was composed of *A. rubrum* and *F. profunda* root-crown sprouts. The growth form of *A. rubrum*, in particular, changes from that of a large, single-stemmed canopy tree where it grows on the top of the natural levee to a multi-stemmed tree where flooding is frequent at the edge of the backswamp. Conner et al. (1981) also reported large densities of *A. rubrum* as a response to flooding in Louisiana impoundments. The consequences of this shift in growth form for the demography of *A. rubrum* or for community dynamics are unknown.

Both *A. rubrum* and *Q. laurifolia* were more common at lower than at upper elevations, despite their classification as only moderately flood tolerant. Moreover, saplings of the exotic tree, *S. sebiferum*, were most common in the lower quadrats, although seedlings were common in gaps throughout the gradient (Denslow and Battaglia, unpub. data). While these distributions, in part, may reflect greater probabilities of establishment on hummocks in frequently flooded areas (Huenneke and Sharitz 1990, Jones 1996, Battaglia et al. 1999), establishment of moderately tolerant species in flood zones may also be affected by canopy openness. Hall and Harcombe (1998) suggest that high light availability may increase flood tolerance in some species. At Lafitte, as in many bottomland hardwood forests, vegetation in the midstory and understory in

frequently flooded sites is less well developed than under better drained conditions (Sharitz and Mitsch 1993), enhancing light availability for seedlings and small saplings. Decline in the abundance of the understory palm, *Sabal minor*, at the lowest elevations may also permit greater penetration of light to the forest floor.

Effects of Hydrologic Change on Regeneration

Of the four moderately tolerant species for which we are able to compare elevation distributions of adults and saplings, two showed an effect of increasing flood frequency and duration associated with subsidence and rising water levels. In contrast to trees of *L. styraciflua* and *U. americana*, saplings were a higher proportion of all saplings at upper than lower elevations. The modal distributions of adults of these species may be as much as 0.5 m lower in elevation compared to saplings, an indication of the magnitude of hydrologic change within the life spans of these trees.

These patterns illustrate flood tolerance differences between adults and saplings, and they suggest that flooding impacts on tree survival may not be a good predictor of flooding impact on tree regeneration. Numerous field and greenhouse studies have shown that seedlings are often particularly vulnerable to flooding (Malecki et al. 1983, Huenneke and Sharitz 1986, 1990, Jones et al. 1989, 1994a, Streng et al. 1989, Conner 1994, King 1995, McLeod et al. 1996, Conner et al. 1998, 1995, Jones and Sharitz 1998) and to flood-

related physical impacts such as scouring and siltation (Huenneke and Sharitz 1990).

The hydrologic changes produced by impoundment are rapid in comparison to those due to subsidence and sea-level rise observed at Lafitte. Species-replacement processes resulting from gradual environmental change may not be easily predictable from patterns observed under rapid change such as that following impoundment. Impoundments have been shown to have detrimental effects on adult trees through reduced growth, crown dieback, increased susceptibility to insects and pathogens, decreased root mass, and increased tree mortality (Harms et al. 1980, Conner et al. 1981, Malecki et al. 1983, Jones et al. 1994b, King 1995, Jones 1996, Keeland et al. 1997). Overstory mortality, in turn, opens the canopy and increases light availability to the understory (Conner et al. 1981). Thus, forest responses to impoundment are the result of both increased flood period and increased light availability to the understory. For example, previously established flood-tolerant, light-demanding species such as *Nyssa aquatica* and *Taxodium distichum* often grow more rapidly in impoundments (Conner et al. 1981, Keeland et al. 1997).

Where relative water-level rise is slower and canopy opening more moderate, as expected from coastal subsidence and/or sea-level rise, growth of flood- and shade-tolerant species such as *A. rubrum* and *F. profunda* may be promoted. We expect that the well-developed sprouting ability of *A. rubrum* will contribute to its dominance in the understory and the suppression of less shade-tolerant species such as *T. distichum* and *N. aquatica*. Regeneration of many bottomland hardwood species may be critically dependent on catastrophic canopy opening due to lightning and windstorms, the frequency of which may also vary as a consequence of global climate change (Pitelka et al. 1997). Harcombe and Marks (1978) note the large number of apparently non-recruiting species in the bottomland hardwood forests of the Big Thicket, Texas. They suggest that regeneration failure may reflect the dependence of these species on major canopy opening events and the influence of understory vegetation on regeneration of canopy trees. The lack of current reproduction in many species at Lafitte likely is related to a combination of factors, including dense understory vegetation and scarce canopy-opening events as well as rising water levels.

The coastal bottomland hardwood forests of the LMAV are being drowned under the gradually rising sea levels produced by a sinking land surface and rising global sea levels. As flood frequency and depth increase, the structure and the composition of the forest will change as a function of the topographic relief imposed by the meander patterns of the old river. Hy-

drologic change in the LMAV is as old as the river itself; it produces the patterns of dynamic structural change that characterize bottomland hardwood forests in contrast to other aspects of the Eastern Deciduous Forest and suggests that the LMAV may be a good model system for the study of the effects of global change. The patterns of forest change may not be straightforward to predict because species differ in their susceptibility to flooding, in the relative effects of flooding on adults, saplings and seedlings, and in light requirements for seedling growth and establishment. Local extinctions and changes in species abundances will reflect both rates of hydrologic change (Williams et al. 1999) and the juxtaposition of other human and natural disturbances, including that due to hurricanes and logging.

Implications for Forest Management

Although trees are often categorized by their light requirements or flood tolerance, species behavior is, in fact, contingent on co-occurring conditions. For example, flood tolerance may depend on the nature of the substrate. This is particularly well-illustrated by differences in composition of bottomland hardwood forests on the sandy soils of the coastal plain and the heavy clay soils of the LMAV (Braun 1950, McKnight et al. 1981, McWilliams and Rosson 1990). Similarly, canopy openness may affect flood tolerance of seedlings and saplings. Seedlings growing in the forest understory are notoriously sensitive to flooding, whereas trees may tolerate an altered hydrologic regime somewhat better. Managers already dealing with a complex forest in a spatially heterogeneous environment, will be challenged with the subtleties of temporal environmental changes occurring over the lifetime of the trees. Understanding and predicting species responses under these circumstances may be improved by recognizing the nature of interactions among species requirements for light and for soil oxygen, nutrient, and moisture supply as modified by effects of their natural enemies and competitors (Jones and Sharitz 1990).

For example, management of these coastal forests for timber production may accelerate shifts in forest composition to more flood-tolerant species rather than promote replacement of canopy trees. In forests subjected to slowly rising water levels (in contrast to the effects of impoundment), we expect that regeneration size classes will become increasingly depauperate in flood-sensitive species while the abundance and vigor of flood-tolerant understory species such as *Sabal minor* and *Acer rubrum* will increase. Seedlings and saplings of overstory species targeted for harvesting may be scarce under adults but more common at higher topographic positions. A rise in the water table follow-

ing timber harvest (Sun et al. 2001) may increase flooding frequency and further diminish regeneration of less flood tolerant species. Without silvicultural treatment, canopy opening produced by a logging operation is likely to promote the further growth of these flood tolerant understory species rather than the regeneration of harvested trees. Maintenance of small-scale topographic heterogeneity in bottomland hardwood forest landscapes will promote seedling establishment of a wide variety of hardwoods and delay somewhat the inevitable flooding of these coastal forests.

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